

Introduction to Machine Learning

What is Machine Learning?

Machine learning is a subfield of artificial intelligence that enables systems to learn patterns from data. Instead of being explicitly programmed with rules, a machine learning model improves its performance on a task as it is exposed to more data.

Supervised Learning

Supervised learning uses labeled training data. Each example in the dataset pairs an input with a known target output. The model learns a mapping from inputs to outputs and can then predict outputs for unseen inputs. Classification and regression are the two main tasks.

Unsupervised Learning

Unsupervised learning finds structure in unlabeled data. Common techniques include clustering, which groups similar examples together, and dimensionality reduction, which compresses data while preserving important structure.

The Bias-Variance Tradeoff

The bias-variance tradeoff describes the tension between a model's ability to fit the training data and its ability to generalize. High bias leads to underfitting, while high variance leads to overfitting.